Scalability & Future Plan

Date	2 July, 2026
Team ID	6a3a4e42cafe96ea01aa3dbb
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	**Limitation**	**Impact**	**Priority to Address(High / Medium / Low)**
1	No caching layer for repeated approval-check queries	Slower response times as user base grows	Medium
2	Single-server deployment with no load balancing	Cannot handle a high volume of concurrent credit card applications	Medium
3	Single-server deployment with no load balancing	Cannot handle a high volume of concurrent credit card applications	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports around 100 concurrent users on a single instance	Deploy on cloud (AWS/Azure) with auto-scaling and load balancers to support 10,000+ concurrent users
2	Data Storage	Local SQL database with limited capacity	Migrate to scalable cloud storage (Amazon RDS / Azure SQL) with partitioning and backups
3	Performance	Synchronous request processing causing delays under load	Introduce asynchronous processing and a Redis caching layer to reduce latency
4	Security	Basic username / password authentication	Implement OAuth 2.0, data encryption in transit/at rest, and role-based access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target timeline	Expected Impact
Phase-2	Real-time integration with credit bureau APIs	04/07/2026	Faster and more accurate approval decisions
Phase-3	Mobile app support and multi-language UI	04/07/2026	Wider user reach and improved accessibility
Phase-4	AI-based fraud detection module	04/07/2026	Enhanced security and reduced fraud risk